Dual-Prompt-Aware Cross-Modal Semantic Interaction and Fusion Network for Remote Sensing Image Captioning

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***Abstract*—Recently, remote sensing image captioning (RSIC) has become an emerging research hot spot that requires models to understand and describe remote sensing images (RSIs). However, the huge modal gap between vision and text makes it difficult to achieve accurate cross-modal transformation for RSIC. Existing methods usually directly transform the vision modal into the text modal based on the multitask learning strategy or visual attention mechanism, which do not make full use of existing prior information to build explicit cross-modal knowledge for vision and text transformation. Considering to utilize the ability of cross-modal alignment in the vision-language model (VLM), we propose a novel dual-prompt-aware cross-modal semantic interaction and fusion network for RSIC. It can explicitly dig out potential entity concepts and predict scene class in the images. And it further builds dual prompts to achieve cross-modal interaction and fusion, which can build a cross-modal common semantic space to provide prior information for caption generation. Specifically, we first introduce an entity-concept exporter (ECE) to obtain explicit entity concepts in the image based on presetting entity space. Next, we design a multiscale scene predictor (MSP) to obtain fine-grained visual semantic features and scene class. Then, we propose a prompt-aware cross-modal interaction (PCI) module to build a cross-modal common semantic space as an intermediate connection for caption generation. Finally, we further design a prompt-aware attention fusion module for the transformer decoder, which can utilize cross-modal prompt features to generate accurate captions. We conduct extensive experiments on three challenging datasets, including UCM-Captions, RSICD, and NWPU-Captions, and our method achieves state-of-the-art (SoTA) performance. In the typical RSIC dataset, RSICD, our method achieves 3.3% and 20.0% improvement in BLEU@4 and CIDEr, respectively, which shows the effectiveness of our method.**

***Index Terms*—Cross-modal interaction, dual prompts, remote sensing image captioning (RSIC), semantic fusion.**

I. Introduction

REMOTE sensing image captioning (RSIC) [1], [2], [3], [4], [5], [6] aims to understand remote sensing image (RSI) and generate the text description in brief corresponding to the scene core content. Although multimodal large language models (MLLMs) have the availability to generate more detailed image descriptions, the traditional RSIC is still valuable for multiple practical applications in real life. First, short captions can provide clear and brief content of RSIs to convey essential information for faster understanding and decision-making. Second, short captions can serve as a form of indexing or tagging for large datasets data management, enabling users to quickly locate relevant images based on core and brief captions. In contrast, longer descriptions generated by MLLMs, while informative and detailed, may not be as efficient for users to quickly understand and query. Third, short captions can be more easily integrated into user interfaces and real-time systems with limited display space, which only conveys the essential and core information and avoids overwhelming the user with longer captions with excessive information. In addition, MLLMs have the advantages of a more detailed caption generation ability, but they are resource-intensive. In contrast, the generation of short captions can be more computationally efficient, which is particularly significant in scenes with limited computational resources or in applications that require quick real-time processing, such as autonomous drones that rely on remote sensing data for navigation and decision-making.

Prior remote sensing captioning methods [6], [7], [8], [9] are usually built based on an encoder–decoder structure with specific-designed attention mechanisms, which uses a convolutional neural network (CNN) to extract visual features in RSIs and uses a recurrent neural network (RNN) to generate text sequence one by one. However, due to the huge modal difference, the transformation from vision modal to text modal is pretty difficult. In addition, RSIs have the characteristic of complex backgrounds, diverse resolution, rich content, dense objects, and diverse object scales, which puts forward higher requirements for the ability of the network to analyze RSIs. To overcome the above challenges in RSIs, as shown in Fig. 1(a), some methods [10], [11], [12], [13], [14], [15] try to learn multiscale visual features and directly fuse them together to avoid incomplete visual representation for global and local objects. Others [5], [16], [17] fully utilize the advantages of multitask learning and predict image scene class as extra auxiliary information to select more important visual features, which can obtain more matching visual context.

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However, most of the above methods just directly transform the vision modal into the text modal, which only implicitly guides the visual feature fusion but does not build an explicit cross-modal connection between vision and text. In other words, they ignore the importance of prior information in the cross-modal common semantic space for cross-modal transformation. This kind of direct transformation might result in the loss of some potential objects in the image, such as buildings and bridges in Fig. 1(a). In all, although these methods have made some progress, it is still hard to achieve outstanding performance in RSIs, especially for more challenging benchmarks with more scene classes and more complex diverse descriptions. Thus, we hope to obtain explicit prompt information to guide cross-modal interaction and fusion, which include the content information about local potential objects and the global topic of the input image, and also have the same modal status as the output caption. It contains enriched content information in a common cross-modal space and can be used to generate more accurate descriptions.

The emergence of the vision-language model (VLM) [18], [19] provides new possibilities for solving the above problems. It has a strong ability of cross-modal alignment between vision and text, which can be used to build explicit prior knowledge to assist in the transformation from vision to text. Considering to utilize the ability of cross-modal alignment in VLM, in this article, we propose a novel dual-prompt-aware cross-modal semantic interaction and fusion network for RSIC, including an entity-concept exporter (ECE), a multiscale scene predictor (MSP), a prompt-aware cross-modal interaction (PCI) module, and a specific-designed transformer decoder with prompt-aware attention fusion module.

As shown in Fig. 1(b), we first design an ECE and an MSP to build dual entity and scene prompts as explicit prior information in text modal. Specifically, the former is built based on the pretrained VLM to exploit potential entity concepts of RSIs in presetting the entity space. The latter digs out fine-grained visual semantic features in different scales and predicts the scene class at the same time. These prompts not only match with the input image, but also contain specific entity concepts and scene information in text modal, such as river and bridge, which are crucial for accurately understanding remote sensing scenes.

Then, considering to explore the common feature space and build a cross-modal semantic connection, we further design the PCI module. Different from the direct transformation from visual features to text or simple cross-modal fusion, our method regards the dual prompts as the explicit visual prior information in text modal, and use them to enhance the prompt-related visual representation in multiscale structure, respectively, which can dig out a cross-modal common semantic space between vision and text as assistance by building the cross-modal interaction between dual prompts and visual features.

Finally, we develop a transformer decoder with prompt-aware attention fusion module, which aims to learn a dynamic cross-modal balance coefficient for each moment based on original visual features, dual-prompt-based cross-modal semantic features, and predicted words to achieve the multiple

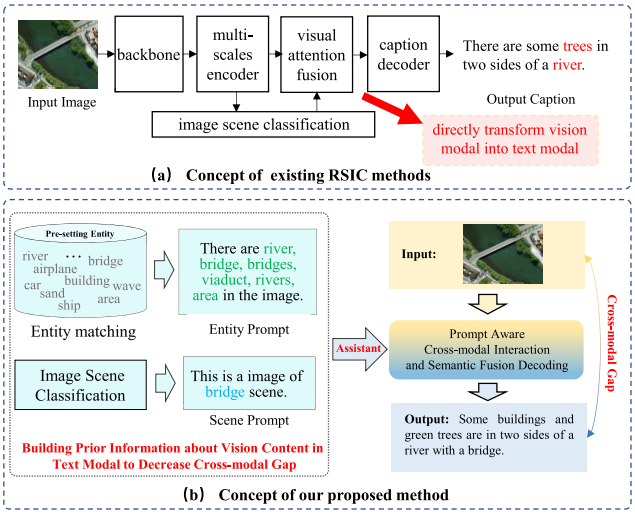


Fig. 1. Comparison of existing RSIC methods and ours. (a) Existing methods aim to directly transform vision modal into text modal with specific-designed attention module or implicit class auxiliary information. (b) We aim to build dual prompts including a scene prompt and an entity prompt, which can provide explicit prior knowledge for caption generation. These prior knowledge not only contain enriched visual content, but also have the status of text modal. We further build PCI and semantic fusion decoding to fully utilize prior information in dual prompts as assistance to decrease the huge cross-modal gap between vision and text, which can assist in caption generation.

semantic adjustments in a common semantic space. It not only considers the influence of vision information and cross-modal prior text prompts, but also considers the particularity of sentence structure and position in the generated captions. These are important for the transformation from vision to text when generating captions. Based on the above dual prompts and specific designed cross-modal interaction and fusion, our method can generate accurate and detailed captions. In summary, our contribution can be summarized as follows.

1) We propose a dual-prompt-aware cross-modal semantic interaction and fusion network, which uses dual prompts based on VLM to build an explicit cross-modal semantic connection and achieve RSIC.
2) We introduce an ECE and an MSP to build dual entity and scene prompts based on VLM and explore the fine-grained visual semantic representation.
3) We further propose a PCI module to build a cross-modal common semantic space based on dual prompts. At the same time, we design a transformer decoder with prompt-aware attention fusion module to achieve precise caption generation.
4) Extensive experiments on the latest and challenging RSIC datasets, including UCM-Captions [7], RSICD [8], and NWPU-Captions [5], show the effectiveness of our method.

It should be noticed that this article is an extended version of our previous work reported in [20]. The main improvements based on the conference version are as follows.

1) We propose a dual-prompt-aware cross-modal semantic interaction and fusion network for RSIC. In addition to using dual-prompt information in text modal, we further design a PCI module to improve the image captioning

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performance. It builds cross-modal semantic connections between vision and text and explores a common semantic space, which takes into account the importance of fine-grained cross-modal semantic interaction before semantic decoding.

2) More recent related works are discussed and compared in this article.
3) More ablation experiments and analysis are presented, including the effectiveness of main modules in our method, the analysis of different prompts, the scales of visual representation, and the influence of pretrained visual and language model.
4) We evaluate our method on various popular and challenging datasets, including UCM-Captions [7], RSICD [8], and NWPU-Captions [5]. These results further demonstrate the effectiveness of our proposed method.

II. Related Work

As a typical cross-modal task in the field of remote sensing, more and more research begin to focus on visual captioning tasks on remote sensing scene, such as RSIC [2], [6], [7], [8], [21], [22], [23] and RSI change captioning [24], [25], [26], [27], [28]. The former aims to generate a detailed text description for the given RSI, while the latter aims to describe the change information between the given image pairs at different times.

Qu et al. [7] first tried to make a semantic-level interpretation for high-resolution RSIs based on the encoder–decoder structure. Shi and Zou [6] proposed a fully convolutional network (FCN) network to consider multilevel image understanding in key instances, envi-elements, and landscapes. Then, Lu et al. [8] built a huge dataset, RSICD, which contains more RSIs and more flexible and complex annotations. Since then, more and more studies have focused on the RSIC task.

Most existing RSIC methods [5], [9], [11], [12], [14], [15], [29], [30] focused on the idea of specific-designed encoder, which aim to explore more sufficient semantic representation from images. Some methods [9], [14], [15] designed different multilevel feature encoders to learn the hierarchical semantic features from RSIs to avoid incomplete visual representation. Li et al. [12] proposed three kinds of attention modules for different regions, different words, and visual semantics. Ma et al. [11] hoped to learn scene-level semantic features and target-level semantic features with better representation based on the designed multiscale attention (MSA) and multifeat attention (MFA) modules. In 2022, a global visual feature-guided attention (GVFGA) mechanism [29] was designed to obtain and fuse the global and local visual features, and a linguistic state (LS) [29] was further used to learn a cross-modal visual–textual attention guidance process. Then, Zhao et al. [30] additionally utilized segmentation proposals to obtain multiple visual features. Cheng et al. [5] proposed a larger and more challenging dataset, NWPU-Captions, and further designed a multilevel and contextual attention network (MLCA-Net), which can adaptively aggregate visual features of different spatial and scales, and explore the latent context information in images. RSCaMa [27] is designed to achieve joint spatial–temporal modeling by building iterative refinement of bitemporal features for more fine-grained visual awareness for images. Liu et al. [25] built a high-quality LEVIR-CC dataset and proposed RSICCFormer to effectively recognize and describe the change details in image pairs.

What is more, to make full use of the advantages of multitask learning, some methods [16], [17], [31], [32], [33] hoped to predict image scene class or topic as extra information and use it help network obtain more matching contextual features to generate detailed description. Zhang et al. [31] proposed a label-attention mechanism (LAM) to predict category information, which was further adopted to guide the masks of the attention module and filter out redundant visual features. Inspired by retrieval tasks, Wang et al. [32] built a topic repository from annotations and used a retrieval strategy to obtain the topic embedding to assist in sentence generation. An instance-awareness and cross-hierarchy attention [33] was proposed with the help of suitable region of interest (ROI) features, including object-level, patch-level, and global visual features, which used the core region proposal network (RPN) and ROIPool modules in detection task. Kandala et al. [16] introduced an auxiliary decoder based on a multilabel scene classification task, which aims to use the conceptual similarity between scene class and caption. In the same year, a joint-training two-stage captioning method (JTTS) [17] was proposed, which was also based on a multilabel classification task and contained a differentiable sampling operator to index the multilabel classification result.

In addition, some researchers also focused on exploring other important impacts, such as training strategies [10], [34] and postprocessing strategies [35]. To extract more fine-grained information from images and enhance the generalization performance of different resolutions, Zhang et al. [10] designed a new training mechanism of multiscale cropping to achieve remote sensing caption. Li et al. [34] also explored the new training strategy and proposed a truncation cross-entropy loss to alleviate the over-fitting problem. BITA [36] designs the learnable prompts to learn visual or text representations for a two-stage caption generation strategy based on VLM and LLM. Hoxha et al. [35] proposed two postprocessing strategies based on hidden Markov models and Viterbi algorithm to avoid the existing error propagation in the recurrent layers.

Recently, the success of transformers, VLM, and meta-learning have attracted much attention from researchers [1], [2], [4], [22], [37], [38], [39]. To exploit the superior performance of self-attention in transformer, CapFormer [22] was designed to use a scalable vision transformer encoder and decoder to achieve a pure transformer caption model. Yang et al. [39] proposed a meta captioning framework to extract meta features from natural image classification and RSI classification. In the same year, Liu et al. [24] proposed a novel method called PromptCC, which creatively decouples the change captioning task into determining whether changes exist and what changes occur, and they further utilized multiprompt learning to introduce large language models (LLMs) to assist in captioning. Liu et al. [28] further built a change-agent, which contains a multilevel visual eye and smart analysis brain, achieving high-quality remote sensing interpretation. In 2024, MVP [2]

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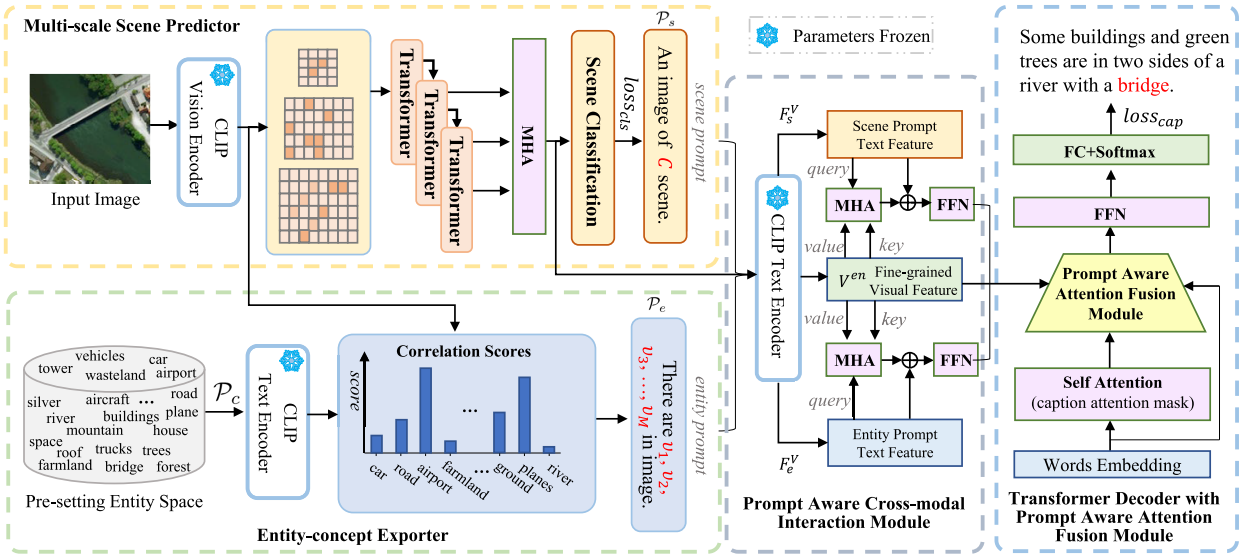


Fig. 2. Framework of our method, which consists of four modules: ECE, MSP, PCI module, and a transformer decoder with prompt-aware attention fusion module. MHA means the multihead attention module, and FFN means the feed-forward network. Better view in color.

aimed to utilize aligned image–text knowledge in VLM and used multiple pretrained feature extractors such as contrastive language-image pre-training (CLIP) and CNN to build a more fine-grained visual representation.

However, most of these RSIC methods directly transform the vision modal into text modal, but do not make full use of the cross-modal alignment ability of VLM and ignore the importance of prior information for cross-modal transformation. Therefore, different from the above methods, we fully dig out the huge cross-modal alignment ability in VLM and exploit various entity concepts and scene classes to build dual prompts based on the CLIP model. It can be used to build an explicit vision–text cross-modal semantic connection, which can help the transformation from vision to text when generating captions.

III. Method

In this section, we show the proposed dual-prompt-aware cross-modal semantic interaction and fusion network for RSIC. The framework of our method is shown in Fig. 2, which includes an ECE, an MSP, a PCI module, and a transformer decoder with prompt-aware attention fusion module. We will introduce the proposed main modules of our method and some details about training objectives in Sections III-A–III-F.

A. Notations

We first introduce some core notations in our method. The goal of our method is to generate a corresponding caption $\mathcal{W} = \{w_1, w_2, \ldots, w_T\}$ for the given RSI $\boldsymbol{I}$. T is the maximum number of words in the generated caption. For the ECE, we aim to extract the top M entity concepts relevant to the input image from the presetting entity space, which is denoted as $\nu = \{\nu_1, \nu_2, \ldots, \nu_M\}$. Based on ν, we build an explicit entity prompt $\mathcal{P}_e$, which contains the entity concepts of the input image. In the MSP, we denote the fine-grained multiscale visual features as $\boldsymbol{V}^v = \{\boldsymbol{V}_1, \boldsymbol{V}_2, \ldots, \boldsymbol{V}_L\}$, and L is the scale number of visual features. After predicting scene class $\mathcal{C}$, we build scene prompt $\mathcal{P}_s$, which contains the scene class clue of the input image.

B. Entity-Concept Exporter

In the RSIs, if only using a simple feature encoder, it is hard to fully dig out the semantic information with rich content and dense objects. To provide more detailed information and build cross-modal prior information for generating vivid captions, we design an ECE to obtain potential entity concepts in the RSIs.

Recently, the pretrained VLM can successfully achieve the alignment between vision and text based on a large amount of vision and text pairs. Thus, we explore the potential entity concepts in the image based on the powerful alignment ability of CLIP. Specifically, we first extract the noun entity concepts in all captions to build the presetting entity space by using the NLTK grammar parser tool.[1] Then, all extracted entity concepts in the space are inserted into a designed prompt $\mathcal{P}_c$ "An image contains {entity concept}." Based on this prompt template, we use the CLIP vision encoder to extract the image representation $\boldsymbol{F_v} \in \mathbb{R}^{1\times C}$ and text representation $\boldsymbol{F_t} \in \mathbb{R}^{E\times C}$, where E is the size of the presetting entity space and C is the size of channel

$$\begin{aligned} \mathcal{P}_c &= \text{“ An image contains entity concept ”} \\ \boldsymbol{F_v} &= \text{CLIP}_v(\boldsymbol{I}) \\ \boldsymbol{F_t} &= \text{CLIP}_t(\mathcal{P}_c) \end{aligned} \tag{1}$$

where CLIP_v and CLIP_t mean the vision encoder and text encoder of CLIP, respectively.

Based on the correlation scores between $\boldsymbol{F_v}$ and $\boldsymbol{F_t}$, we select the M entity concepts $\nu = \{\nu_1, \nu_2, \ldots, \nu_M\}$ with the

[1] https://www.nltk.org/

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highest scores. Finally, to obtain the fine-grained entity information, we build the explicit entity prompt template $\mathcal{P}_e$

$$\mathcal{P}_e = \text{“ There are } \nu_1, \ldots, \nu_M \text{ in the image ”} \quad (2)$$

where ν_m is the mth related entity concept of the input image.

C. Multiscale Scene Predictor

Due to the typical characteristics of RSIs such as diverse resolution and varied viewpoints, just using single visual feature is insufficient to represent visual and semantic information of objects with different scales and shapes. Thus, we consider learning a multiscale visual representation to solve the above problems. In addition, we further use the fine-grained visual features to predict scene class to build scene prompt information, which can provide more explicit prior information in the text modal as a connection between vision and text.

As for the image representation $\boldsymbol{F}'_v \in \mathbb{R}^{K_v \times C}$ obtained from pretrained CLIP, we downsample $\boldsymbol{F}'_v$ for L times to obtain the initial multiscale visual features $\boldsymbol{V}^v = \{\boldsymbol{V}_1, \boldsymbol{V}_2, \ldots, \boldsymbol{V}_L\}$. $\boldsymbol{F}'_v$ is the output of last layer before $\boldsymbol{F}_v$ in (1). We jointly optimize visual features with different scales based on the scene class supervision to achieve fine-grained multiscale feature extraction.

We first stack N_e basic transformer blocks with multihead attention (MHA) in total for multiscale feature encoding to further extract high-level visual semantic features in $\boldsymbol{V}^v$. For example, for the ith transformer block and lth visual feature, the operation can be expressed as follows:

$$\begin{aligned}
\boldsymbol{V}_l^{'i} &= \text{LN}(\text{Dropout}(\text{MHA}(\boldsymbol{V}_l^{i-1}, \boldsymbol{V}_l^{i-1}, \boldsymbol{V}_l^{i-1})) + \boldsymbol{V}_l^{i-1}) \\
\boldsymbol{V}_l^{''i} &= \text{Dropout}(\text{ReLU}(\boldsymbol{V}_l^{'i}\boldsymbol{W}_1)) \\
\boldsymbol{V}_l^{i} &= \text{LN}(\text{Dropout}(\boldsymbol{V}_l^{''i}\boldsymbol{W}_2) + \boldsymbol{V}_l^{'i})
\end{aligned} \quad (3)$$

where $\boldsymbol{W}_*$ represents the learnable parameter matrices, and $\boldsymbol{V}_l^i$ means the output of ith transformer block for lth visual feature, $\boldsymbol{V}_l^i \in \mathbb{R}^{K_v \times C}$. MHA represents a multihead attention module, rectified linear unit (ReLU) represents a nonlinear activation function, Dropout represents the random deactivation, and LN represents the layer normalization.

Then, we use an average pooling operation to extract global context information $\boldsymbol{G}_l$ from the highest-level visual information $\boldsymbol{V}_l^{N_e}$. Then, we save and merge all global context information based on a fully connected layer

$$\boldsymbol{G} = \text{FC}(\text{Cascade}(\boldsymbol{G}_1, .., \boldsymbol{G}_L)) \quad (4)$$

where $\text{Cascade}(*, *)$ means the feature cascading operation, $\boldsymbol{G}_l \in \mathbb{R}^{1 \times C}$, and $\boldsymbol{G} \in \mathbb{R}^{1 \times C}$.

After passing all transformer encoder blocks, we get the global visual feature $\boldsymbol{G}$ and fine-grained multiscale visual features $\boldsymbol{V}^{\text{en}} = \{\boldsymbol{V}_1^{N_e}, \ldots, \boldsymbol{V}_L^{N_e}\}$, which provide more visual context information for decoding. We further use the global visual feature $\boldsymbol{G}$ to predict the scene class $\mathcal{C}$ as another extra cross-modal prior information for caption generation

$$\boldsymbol{P}_{\text{cls}} = \text{Softmax}(\text{FC}(\text{Dropout}(\boldsymbol{G}))) \quad (5)$$

where $\boldsymbol{P}_{\text{cls}}$ is the probability distribution of the scene class.

Finally, we insert the scene class result $\mathcal{C}$ into a scene prompt template $\mathcal{P}_s$

$$\mathcal{P}_s = \text{“ An image of } \mathcal{C} \text{ scene ”} \quad (6)$$

where $\mathcal{C}$ is the predicted scene class result.

D. PCI Module

Based on the above modules, we build the dual prompts and introduce explicit prior information in the text modal, which can provide more information for caption generation and build an explicit connection between vision and text. As is known to all, there is obvious gap between vision modal and text modal, which leads to poor performance in direct cross-modal transformation. Based on the designed dual prompts, we hope to use explicit prompt connection in text modal to drive the cross-modal semantic interaction, which can dig out a cross-modal common semantic space between vision and text to assist in the transformation from fine-grained visual features into text modal space.

Specifically, with the help of the ECE and scene class prediction, the extracted entities ν and scene class $\mathcal{C}$ have been inserted into the prompt templates $\mathcal{P}_e$ and $\mathcal{P}_s$. Thus, we use the CLIP text encoder to extract text representation $\boldsymbol{F_e}$ and $\boldsymbol{F_s}$ of $\mathcal{P}_e$ and $\mathcal{P}_s$, which not only match with the input image, but also contain explicit semantic information in text modal

$$\boldsymbol{F_e} = \text{CLIP}_t(\mathcal{P}_e) \quad \boldsymbol{F_s} = \text{CLIP}_t(\mathcal{P}_s) \quad (7)$$

where $\boldsymbol{F_e} \in \mathbb{R}^{K_e \times C}$ and $\boldsymbol{F_s} \in \mathbb{R}^{K_s \times C}$.

Then, we design a cross-modal interaction module with a transformer structure to explore the core semantic information in the fine-grained visual feature $\boldsymbol{V}^{\text{en}}$ based on multiple prompts. It can align and interact $\boldsymbol{V}^{\text{en}}$ with the dual-prompt prior information, and select the effective visual information into the cross-modal common feature space. As for the core visual information corresponding to the scene class, we use $\boldsymbol{F_s}$ as query to select the fine-grained visual feature $\boldsymbol{V}^{\text{en}}$ to obtain the cross-modal scene prompt semantic feature $\boldsymbol{F}_s^{\text{cross}}$

$$\begin{aligned}
\boldsymbol{F}'_s &= \text{LN}(\text{Dropout}(\text{MHA}(\boldsymbol{F_s}, \boldsymbol{V}^{\text{en}}, \boldsymbol{V}^{\text{en}})) + \boldsymbol{F_s}) \\
\boldsymbol{F}''_s &= \text{Dropout}(\text{ReLU}(\boldsymbol{F}'_s\boldsymbol{W}_1)) \\
\boldsymbol{F}_s^{\text{cross}} &= \text{LN}(\text{Dropout}(\boldsymbol{F}''_s\boldsymbol{W}_2) + \boldsymbol{F}'_s)
\end{aligned} \quad (8)$$

where $\boldsymbol{F_s}$ is the text representation of scene prompt, $\boldsymbol{V}^{\text{en}}$ is the fine-grained visual feature, and $\boldsymbol{F}_s^{\text{cross}}$ is the cross-modal scene prompt semantic feature. $\boldsymbol{F_s} \in \mathbb{R}^{K_s \times C}$ and $\boldsymbol{F}_s^{\text{cross}} \in \mathbb{R}^{K_s \times C}$.

For the entity prompt $\boldsymbol{F_e}$, we also extract the entity prompt semantic feature $\boldsymbol{F}_e^{\text{cross}}$ by using a similar operation. These two kinds of prompt-based cross-modal semantic features contain enriched prior information about the image entity concepts and scene class for caption generation.

E. Transformer Decoder With Prompt-Aware Attention Fusion Module

Based on the two kinds of prompt-based cross-modal semantic features $\boldsymbol{F}_e^{\text{cross}}$ and $\boldsymbol{F}_s^{\text{cross}}$, we develop a specific-designed transformer decoder with prompt-aware attention fusion module, which contains N_d transformer blocks and

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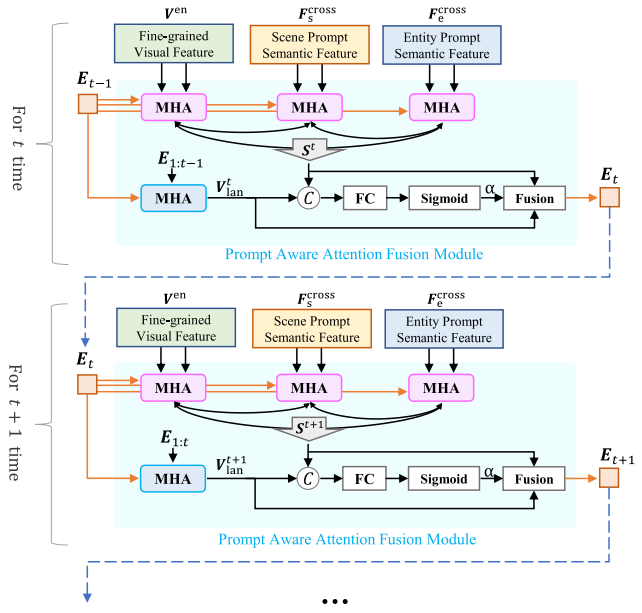


Fig. 3. Structure of prompt-aware attention fusion module in our transformer decoder on t time and $t+1$ time, which is used to obtain the hidden state of different moments for word prediction. Our method generates caption word by word, which generates the word at the current time based on the last time. At t time, $\boldsymbol{E}_{t-1}$ means the word embedding at the last time. The MHA is the multihead attention module. The pink MHAs are used to select the semantic representation related to the current time in fine-grained visual features and dual-prompt semantic features. The black double arrow lines mean building the cross-modal semantic interaction to obtain the fused semantic $\boldsymbol{S}^t$. The blue MHA is used to learn the sentence structure information and textual semantic information $\boldsymbol{V}_{\text{lan}}^t$. Finally, our method learns the balance coefficient α to fuse the cross-modal feature and sentence information, which can be used to predict the word at the current time.

can achieve dynamic cross-modal feature fusion and accurate caption generation.

As shown in Fig. 3, different from existing transformer structure just using a cross-modal attention to fuse vision and text features to generate caption, we design a prompt-aware attention fusion module, which can learn a dynamic cross-modal balance coefficient at different moments based on dual-prompt-based cross-modal semantic features and predicted words to achieve the multiple semantic adjustment in common semantic space.

As is known to all, when predicting words at different moments, not all semantic information is useful, which greatly depends on the text state at the current moment. Thus, we use the predicted word embedding feature $\boldsymbol{E}_{t-1}$ at the last moment to select and merge important semantic information in the dual-prompt-based cross-modal semantic features and fine-grained visual feature based on an MHA mechanism

$$\begin{aligned}\boldsymbol{S}_v^t &= \text{MHA}(\boldsymbol{E}_{t-1}, \boldsymbol{V}^{\text{en}}, \boldsymbol{V}^{\text{en}})\\ \boldsymbol{S}_s^t &= \text{MHA}(\boldsymbol{E}_{t-1}, \boldsymbol{F}_s^{\text{cross}}, \boldsymbol{F}_s^{\text{cross}})\\ \boldsymbol{S}_e^t &= \text{MHA}(\boldsymbol{E}_{t-1}, \boldsymbol{F}_e^{\text{cross}}, \boldsymbol{F}_e^{\text{cross}})\end{aligned} \tag{9}$$

where $\boldsymbol{E}_{t-1}$ is the word embedding feature at $t-1$ time, $\boldsymbol{E}_{t-1} \in \mathbb{R}^{1\times C}$, and $\boldsymbol{S}_*^t \in \mathbb{R}^{1\times C}$.

Then, we use a self-attention mechanism to merge the above adjusted semantic features $\boldsymbol{S}_v^t$, $\boldsymbol{S}_e^t$, and $\boldsymbol{S}_s^t$

$$\boldsymbol{S}^t = \text{S-Att}(\text{Cascade}(\boldsymbol{S}_v^t, \boldsymbol{S}_e^t, \boldsymbol{S}_s^t)) \tag{10}$$

where $\boldsymbol{S}^t$ is the fused cross-modal semantic feature and $\boldsymbol{S}^t \in \mathbb{R}^{1\times C}$. S-Att means the self-attention mechanism.

With time going by, more and more words are predicted. Therefore, sentence structure information and textual semantic information are dynamically changing. We also use an MHA mechanism to enhance text feature $\boldsymbol{E}_{1:t-1}$, which is composed of the word embedding of all time before the current moment

$$\boldsymbol{V}_{\text{lan}}^t = \text{MHA}(\boldsymbol{E}_{t-1}, \boldsymbol{E}_{1:t-1}, \boldsymbol{E}_{1:t-1}) \tag{11}$$

where $\boldsymbol{E}_{t-1} \in \mathbb{R}^{1\times C}$, $\boldsymbol{E}_{1:t-1} \in \mathbb{R}^{(t-1)\times C}$, and $\boldsymbol{V}_{\text{lan}}^t \in \mathbb{R}^{1\times C}$.

Based on the fused cross-modal semantic feature $\boldsymbol{S}^t$ and the generated language feature $\boldsymbol{V}_{\text{lan}}^t$, we learn the balance coefficient $\boldsymbol{\alpha}$ of them at the current moment to achieve the semantic features fusion. After obtaining the fused cross-modal feature $\boldsymbol{V}_t^{\text{cm}}$, we use the feed-forward layer in the transformer block to learn a hidden state in the text feature space

$$\begin{aligned}\boldsymbol{\alpha} &= \text{Sigmoid}(\text{Cascade}(\boldsymbol{V}_{\text{lan}}^t, \boldsymbol{S}^t)\boldsymbol{W}_1)\\ \boldsymbol{V}_t^{\text{cm}} &= \boldsymbol{\alpha}\boldsymbol{V}_{\text{lan}}^t + (1-\boldsymbol{\alpha})\boldsymbol{S}^t\\ \boldsymbol{H}_t' &= \text{Dropout}(\text{ReLU}(\boldsymbol{V}_t^{\text{cm}}\boldsymbol{W}_2 + \boldsymbol{H}_{t-1}))\\ \boldsymbol{H}_t &= \text{LN}(\text{Dropout}(\boldsymbol{H}_t'\boldsymbol{W}_3) + \boldsymbol{H}_t')\end{aligned} \tag{12}$$

where $\boldsymbol{\alpha}$ is the learnable balance coefficient for the fused cross-modal semantic feature $\boldsymbol{S}^t$ and the generated language feature $\boldsymbol{V}_{\text{lan}}^t$. $\boldsymbol{\alpha} \in \mathbb{R}^{1\times C}$. $\boldsymbol{H}_{t-1}$ means the hidden state at $t-1$ time of the last transformer block. Sigmoid and ReLU represent nonlinear activation functions, Cascade represents the feature cascading operation, Dropout represents the random deactivation, and LN represents the layer normalization.

At the end, we use a fully connected layer to predict word w_t at t time based on the hidden state $\boldsymbol{H}_t$ of the last transformer block

$$\boldsymbol{P}_{w_t} = \text{Softmax}(\text{FC}(\text{Dropout}(\boldsymbol{H}_t))) \tag{13}$$

where $\boldsymbol{P}_{w_t}$ is the probability distribution of the word at t time.

F. Objectives

Except for the pretrained vision encoder and text encoder in VLM, we jointly train all proposed modules based on the scene classification task and caption generation task.

For the scene classification task, we use the cross-entropy loss as the objective function

$$\mathcal{L}_{\text{cls}} = -\log \boldsymbol{P}_{\mathbf{cls}}\boldsymbol{Y}_{\mathbf{cls}}^{\top} \tag{14}$$

where $\boldsymbol{Y}_{\mathbf{cls}}$ means the one-hot vector of the ground truth (GT) about scene class, and $\top$ means the matrix transposition.

For the caption generation task, according to the prediction probability of the word w_t, we also use the common cross-entropy loss in captioning task to constrain words generation for training stage

$$\mathcal{L}_{\text{cap}} = -\sum_{t=1}^{T} \log \boldsymbol{P}_{w_t}\boldsymbol{Y}_{w_t}^{\top} \tag{15}$$

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TABLE I
DATA STATISTICS OF REMOTE SCENE IMAGE CAPTIONING DATASETS

Datasets	images	sentence	sentence per image	average sentence length	vocabulary size	scene classes number
UCM-Captions [7]	2100	10500	5	10.5	280	21
RSICD [8]	10921	24333 / 54605	1-5 / 5	10.6	1169	30
NWPU-Captions [5]	31500	157500	5	12.3	1462	45
VRSBench [41]	29614	29614	1	54	1849	w/o scene annotation

where T is the maximum length of captions. $\boldsymbol{Y}_{w_t}$ means the one-hot vector of the GT about the word at time t.

Thus, the final objective function is as follows:

$$\mathcal{L} = \beta \cdot \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{cap}} \tag{16}$$

where β is the balance parameter of multitask learning.

IV. EXPERIMENTS

We conduct massive experiments based on three public datasets, including UCM-Captions [7], RSICD [8], and NWPU-Captions [5] to evaluate the effectiveness of our method. More details and results about our experiments will be introduced in this section.

A. Datasets

1) UCM-Captions: In 2016, UCM-Captions [7] was built on UC Merced land-use dataset [41], which is the earliest remote sensing image captioning dataset. It contains 2100 aerial images with 21 scene classes, in which the size of all images is 256 × 256 pixels. Each scene class has 100 images. For each image, it is annotated with five captions as the description.

2) RSICD: RSICD [8] is built on different image sources, including Baidu Map, Google Earth, MapABC, and Tianditu, which contains 10 921 images with 30 scene classes. The size of images is 224 × 224 pixels. Different from UCM-Captions, each image in RSICD has a different number of caption annotations. The original dataset only contains 24 333 captions in total, and then, Lu et al. [8] use a randomly duplicating strategy to extend the number of captions to 54 605 and ensure each image has five captions.

3) NWPU-Captions: NWPU-Captions [5] is built on the NWPU-RESISC45 dataset in 2022, which is one of the largest public remote sensing image captioning datasets. It contains 31 500 images with 45 scene classes, which are collected from Google Earth with a size of 256 × 256 pixels. Each scene class has 700 images. Similar to UCM-Captions, each image is annotated manually with five different captions as the description.

4) VRSBench: VRSBench [40] was published in 2024, which is one of the latest remote sensing datasets. It contains 29 614 images, and each image is manually annotated with one detailed caption, which provides comprehensive descriptions about both abstract image attributes and detailed object-specific information. Each caption typically comprises three–seven sentences, with an average length of 54 words.

We further comprehensively compare the above four public datasets of RSIC, and the detailed data statistics are shown in Table I. It can be seen that RSICD, NWPU-Captions, and VRSBench are more challenging, which have more images, more scene classes, or more detailed captions, and the caption annotations contain more vocabularies, which are more diverse and complex.

B. Metrics

To comprehensively evaluate the quality of generated captions, we use four kinds of common single language score and one mean language score to build the evaluation metrics of our method. Specifically, the single language score includes BLEU@1–4 (B@1–4) [42], METEOR (M) [43], ROUGE_L (R) [44], and CIDEr (C) [45].

1) BLEU@n mainly calculates the score based on the overlapping n-gram words between the generated caption and GT captions.
2) METEOR measures the accuracy of captions by introducing an external dictionary, which considers the influence of synonyms.
3) ROUGE_L calculates the recall of generated captions and GT captions based on the length of the longest common subsequence.
4) CIDEr makes a word frequency statistics to measure the diversity and semantic consistency of generated captions.

Due to the multiple evaluation metrics for evaluating the image captioning models, AI challenge[2] designs a mean language score S_m to overall consider multiple evaluation metrics in 2017

$$S_m = \frac{\text{B@4} + \text{M} + \text{R} + \text{C}}{4}. \tag{17}$$

C. Implementation Details

In our method, the encoders of vision and text are initialized by pretrained RemoteCLIP-B-32 [18], [19]. In the other proposed modules in our method, we adopt the random initialization. For the ECE, we set the M as 6 for UCM-Captions, RSICD, and NWPU-Captions in entity concepts $v = \{v_1, v_2, \ldots, v_M\}$ with the highest scores. For the MSP, we downsample visual features for 3× and stack $L_e = 3$ transformer blocks. For the processing of vision and text embedding, the dimensions of the embedding layers are both 512. In the decoding stage, we stack $L_d = 3$ prompt-based transformer blocks, and the hidden size in the decoder is also set to 512. For MHA(Q, K, V), the head number is 8.

For the word tokenizer, we use a learnable word vector group, whose size is based on the vocabulary list. We build

[2]https://challenger.ai/competition/caption

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a vocabulary list based on caption annotations and remove all stop words and punctuation. In addition, the low-frequency words that appear less than 5× are also removed. Finally, the sizes of the vocabulary lists for UCM-Captions, RSICD, and NWPU-Captions are 281, 1170, and 1463, respectively. At the same time, we use natural language processing (NLP) tools to analyze the part of speech, and then construct presetting entity space based on the noun. The sizes of presetting entity space for them are 75, 163, and 439, respectively. In addition, we change all words to lowercase, stripped of extra space in the head or end, and add <eos> in the end as an ending symbol.

All experiments are conducted based on Xmodaler [46] toolkit. During the training stage, our method is optimized based on AdamW, and the initial learning rate is $4e^{-4}$. We freeze all parameters of the vision encoder and text encoder in CLIP. We use the step decay mechanism with a warm-up strategy. The size of the warm-up strategy is three training epochs. The learning rate decreases once every three training epochs, and the decay weight is 0.6. The balance parameter β in the loss function is set to 0.5. For UCM-Captions, RSICD, and NWPU-Captions, the max length of caption is 23, 25, and 27, respectively. The training epoch is 12. During the inference stage, we use a beam search strategy. The size of the beam search is set to 3.

TABLE II

ABLATION STUDY OF MAIN MODULES ON RSICD AND NWPU-CAPTIONS. PCI MEANS PROMPT-AWARE CROSS-MODAL INTERACTION MODULE, AND PAF-T MEANS TRANSFORMER DECODER WITH PROMPT-AWARE ATTENTION FUSION MODULE

Datasets	Main Modules		B@1	B@2	B@3	B@4	M	R	C	S_m
	PCI	PAF-T								
RSICD	Baseline		81.3	70.8	61.8	54.2	39.3	70.8	299.0	115.8
		✓	81.7	71.8	63.2	55.8	39.8	71.7	304.7	118.0
	✓		82.6	72.2	63.6	56.4	40.7	72.1	309.4	119.7
	✓	✓	**82.9**	**72.8**	**64.2**	**57.0**	**40.8**	**72.1**	**312.5**	**120.6**
NWPU-Captions	Baseline		90.1	82.5	75.9	70.3	44.1	78.4	202.4	98.8
		✓	89.9	82.8	76.6	71.3	44.9	79.0	205.1	100.1
	✓		**90.9**	84.1	78.3	73.4	45.8	79.9	209.8	102.2
	✓	✓	**90.9**	**84.2**	**78.6**	**73.8**	47.2	**81.4**	**214.1**	**104.1**

TABLE III

ABLATION STUDY OF THE EFFECTIVENESS OF SCENE PROMPT $\mathcal{P}_s$ AND ENTITY PROMPT $\mathcal{P}_e$ ON RSICD AND NWPU-CAPTIONS

Datasets	$\mathcal{P}_s$	$\mathcal{P}_e$	B@1	B@2	B@3	B@4	M	R	C	S_m
RSICD	✓		82.1	72.0	63.4	56.0	40.5	71.5	302.9	117.7
		✓	82.1	71.7	62.9	55.3	40.4	71.5	306.5	118.4
	✓	✓	**82.9**	**72.8**	**64.2**	**57.0**	**40.8**	**72.1**	**312.5**	**120.6**
NWPU-Captions	✓		90.4	83.3	77.3	72.2	46.3	80.4	207.6	101.6
		✓	90.3	83.6	77.9	73.0	46.4	80.4	209.0	102.2
	✓	✓	**90.9**	**84.2**	**78.6**	**73.8**	47.2	**81.4**	**214.1**	**104.1**

D. Ablation Study

In this section, we conduct massive ablation studies to analyze the effectiveness of our proposed model on the RSICD and NWPU-Captions. First, we explore the effectiveness of the main modules. Second, we verify the influence of entity number when building entity prompt $\mathcal{P}_e$ in the ECE. Third, we study the number of feature scales in a multi-scale structure. Next, we explore the effectiveness of scene prompt $\mathcal{P}_s$ and entity prompt $\mathcal{P}_e$ in our method. Finally, we explore the importance of multiple pretrained CLIP model for RSIC.

1) Effectiveness of Main Modules: Our proposed method mainly contains four core modules, including an ECE, an MSP, a PCI module, and a transformer decoder with prompt-aware attention fusion (PAF-T) module. It should be noted that PCI and PAF-T rely on the dual prompts in ECE and MSP. Thus, in this section, we keep the dual prompts to make ablation studies of modules PCI and PAF-T. We will explore the impact of dual prompts individually in the next part. The experimental results are shown in Table II. It should be noticed that the baseline method is a simple transformer-based encoder–decoder structure.

It can be seen that introducing the dual-prompt prior information into the transformer decoder with prompt-aware attention fusion module (PAF-T) can improve the quality of the caption to a certain extent. The average score S_m has been improved from 115.8 to 118.0 and 98.8 to 100.1 on RSICD and NWPU-Captions, respectively. The reason is that more prior information in text modal are of vital importance for the decoding stage, which can improve the visual representation ability and explicitly provide more semantic information in the image for caption generation. When individually using the PCI module to explore the common semantic space and build cross-modal semantic connections, the improvement is pretty significant. This is because the explicit prior information from scene prompt $\mathcal{P}_s$ and entity prompt $\mathcal{P}_e$ can dig out more important fine-grained semantic representations by cross-modal interaction instead of using fixed visual features for decoding.

It can be seen that "PCI + PAF-T" achieves the best performance of the caption generation, especially the CIDEr score. When using the dual-prompt prior information to guide cross-modal interaction, it can build a cross-modal common feature space and provide a cross-modal semantic connection to narrow the distance between vision and text. After obtaining the high-quality cross-modal semantic information, the transformer decoder with prompt-aware attention fusion (PAF-T) module can dynamically achieve multiple semantic adjustment and fusion based on the generated words. Thus, the combination of them can observably improve the quality of the generated caption. The experimental results fully demonstrate the effectiveness of our method.

2) Effectiveness of Two Kinds of Prompts: In our method, we introduce two kinds of explicit prompts, including a scene prompt $\mathcal{P}_s$ and an entity prompt $\mathcal{P}_e$. We make detailed ablation studies to explore the influence of two kinds of prompts, and the experimental results are shown in Table III. It can be seen that introducing either prompt alone significantly improves the quality of the generated caption, and the combination of the two prompts achieves the best performance. Since scene prompt can be obtained under the condition of scene class supervision, $\mathcal{P}_s$ can provide accurate scene prior information for caption generation. For $\mathcal{P}_e$, it is obtained through the matching of vision and text based on the pretrained CLIP, which contains more enriched prior information and can provide multiple potential entity concepts as cross-modal

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TABLE IV

ABLATION STUDY OF ENTITY NUMBER IN ECE ON RSICD AND NWPU-CAPTIONS DATASETS

Datasets	Entity Number	B@1	B@2	B@3	B@4	M	R	C	S_m
RSICD	4	81.7	71.7	63.1	55.8	40.1	71.4	307.5	118.7
	5	82.5	72.0	63.2	55.8	40.6	71.9	310.6	119.7
	6	**82.9**	**72.8**	**64.2**	**57.0**	**40.8**	**72.1**	**312.5**	**120.6**
	7	81.7	71.5	63.1	55.8	40.1	71.7	303.6	117.8
	8	81.0	70.6	62.0	54.6	40.0	71.1	299.8	116.4
NWPU-Captions	4	90.2	83.2	77.2	72.2	46.2	79.9	207.6	101.5
	5	90.8	83.9	78.0	73.0	46.1	80.5	211.0	102.7
	6	**90.9**	**84.2**	**78.6**	**73.8**	47.2	**81.4**	**214.1**	**104.1**
	7	90.7	84.0	78.2	73.3	46.7	80.9	210.0	102.7
	8	90.0	83.1	77.3	72.4	46.3	80.4	208.9	102.0

TABLE V

ABLATION STUDY OF SCALE NUMBER FOR MULTISCALE STRUCTURE ON RSICD AND NWPU-CAPTIONS

Datasets	Scale Number	B@1	B@2	B@3	B@4	M	R	C	S_m
RSICD	1	81.9	72.0	63.4	55.8	40.2	71.7	308.4	119.0
	2	82.4	72.2	63.6	56.2	40.5	72.0	309.0	119.4
	3	**82.9**	**72.8**	**64.2**	**57.0**	**40.8**	**72.1**	**312.5**	**120.6**
NWPU-Captions	1	90.2	83.6	77.9	73.0	46.2	80.6	208.1	102.0
	2	90.8	84.0	78.3	73.4	46.5	80.6	211.4	102.9
	3	**90.9**	**84.2**	**78.6**	**73.8**	47.2	**81.4**	**214.1**	**104.1**

connections to assist in word prediction. Thus, the above prompts can improve the performance of RSIC.

3) Effectiveness of Entity Number in ECE: In the ECE, the key parameter is the number of entity concepts, which determines the amount of information contained in $\mathcal{P}_e$. Thus, we set different entity numbers and evaluate the quality of caption generation. As shown in Table IV, when the entity number increases from 4 to 6, almost all metrics of caption are improved gradually. However, when the entity number is too large, the performance has significantly decreased. We analyze that when the entity number is small, it can effectively provide prior information to assist in caption generation, and when the entity number is too large, it will introduce some noise and error entity concepts, which affects the accuracy of decoding. Thus, in our final method, we set the entity number to 6.

4) Effectiveness of Scale Number in Multiscale Structure: Further, we explore the multiscale structure to extract fine-grained visual representation. We set the scale number to 1, 2, and 3 for the ablation study, and the experimental results are shown in Table V. The results indicate that introducing multiscale visual features can obtain more fine-grained semantic information in the image and improve the performance of our model. In addition, with the explicit driving of scene prompt and entity prompt, fine-grained visual semantic features can further improve the quality of cross-modal semantic connection and provide more detailed semantic information for caption generation.

5) Effectiveness of Pretrained CLIP Models: Finally, we use three kinds of pretrained CLIP models to make ablation studies, including random initialization, initialization with CLIP Vit-B-32 [18], initialization with RemoteCLIP Vit-B-32 [19], and initialization with Git-RSCLIP [47]. The

TABLE VI

ABLATION STUDY OF DIFFERENT FEATURE EXTRACTION METHODS ON RSICD AND NWPU-CAPTIONS. UNDER THE RANDOM INITIALIZATION SETTING, WE WILL NOT LOAD ANY PARAMETERS FOR THE VISION OR TEXT ENCODER

Datasets	Methods	B@1	B@2	B@3	B@4	M	R	C	S_m
RSICD	Random Initialization	79.5	68.5	59.5	52.1	38.7	68.9	289.8	112.4
	CLIP ViT-B-32 [18]	82.3	72.1	63.6	56.4	40.1	71.7	**312.6**	120.2
	RemoteCLIP ViT-B-32 [19]	**82.9**	72.8	64.2	57.0	**40.8**	72.1	312.5	120.6
	Git-RSCLIP [47]	**82.9**	**73.1**	**64.6**	**57.2**	40.7	**72.4**	**312.6**	**120.7**
NWPU-Captions	Random Initialization	86.7	77.9	70.8	64.9	40.4	74.1	184.2	90.9
	CLIP ViT-B-32 [18]	90.3	83.3	77.4	72.5	46.2	80.3	206.5	101.4
	RemoteCLIP ViT-B-32 [19]	90.9	84.2	78.6	73.8	47.2	81.4	214.1	104.1
	Git-RSCLIP [47]	**91.9**	**85.6**	**80.1**	**75.4**	**49.9**	**82.4**	**216.7**	**106.1**

TABLE VII

EXPERIMENTAL RESULTS ABOUT WHETHER TO FREEZE OR FINE-TUNE THE PRETRAINED CLIP MODEL ON RSICD AND NWPU-CAPTIONS. VT LAYER MEANS VISION AND TEXT LAYER

Datasets	Training Mode	B@1	B@2	B@3	B@4	M	R	C	S_m
RSICD	Ours (Frozen All VT Layers)	**82.9**	**72.8**	**64.2**	**57.0**	**40.8**	72.1	**312.5**	**120.6**
	Fine-tune Last 1 VT Layer	82.4	72.4	63.9	56.7	40.5	**72.3**	311.3	120.2
	Fine-tune Last 2 VT Layers	82.3	72.0	63.1	55.5	40.2	71.7	307.9	118.8
	Fine-tune All Vision Layers	80.9	70.8	62.6	55.6	40.1	71.5	301.3	117.1
	Fine-tune All Text layers	82.0	71.7	63.0	55.5	39.8	71.5	306.7	118.4
	Fine-tune All VT Layers	81.4	70.7	61.8	54.4	40.5	71.6	298.0	116.1
NWPU-Captions	Ours (Frozen All VT Layers)	**90.9**	**84.2**	**78.6**	**73.8**	**47.2**	**81.4**	**214.1**	**104.1**
	Fine-tune Last 1 VT Layer	90.7	83.9	78.1	73.1	46.3	80.4	211.5	102.8
	Fine-tune Last 2 VT Layers	90.6	83.7	77.8	72.8	46.0	80.2	210.2	102.3
	Fine-tune All Vision Layers	90.5	83.7	77.7	72.6	46.1	80.3	207.1	101.5
	Fine-tune All Text layers	90.5	83.5	77.7	72.7	46.3	80.4	210.3	102.4
	Fine-tune All VT Layers	89.6	82.5	76.4	71.2	46.0	80.1	205.5	100.7

experimental results are shown in Table VI. It can be seen that the CIDEr and S_m in the random initialization setting can arrive at 289.9% and 112.4%, respectively. In this kind of setting, the cross-modal alignment in pretrained VLM cannot be utilized. When using any kind of pretrained models, the quality of the generated caption achieves huge improvement, which should be attributed to the cross-modal alignment ability brought by pretrained models. In addition, pretrained CLIP models [19], [47] for RSIs have achieved better performance compared to general pretrained CLIP [18], and by using the latest Git-RSCLIP [47] pretrained on the Git-10M dataset, our method achieves the best caption performance on almost all metrics. It should be noticed that for other experimental results in this article, we use the pretrained model of RemoteCLIP ViT-B-32 as the vision and text encoder.

In addition, we further explore the training strategy for whether to freeze or fine-tune the pretrained CLIP model. Fine-tuning experiments are achieved based on the end-to-end structure with the weight coefficient of learning rate 0.1 for the fine-tuned layers. As shown in Table VII, freezing the CLIP model achieved the best experimental performance on both RSICD and NWPU-Captions datasets. Even if only the last one or two layers are fine-tuned, the overall performance still shows a decline. In the case of fine-tuning all vision or text layers, the experimental performance drops obviously. We think freezing the pretrained parameters can preserve these general features with the generalization ability, benefiting from multiple large and diverse datasets. In addition, compared with the massive datasets for pretraining, the size of the remote sensing caption dataset might be relatively smaller. Fine-tuning

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TABLE VIII
EXPERIMENTAL RESULTS ON UCM-CAPTIONS DATASET

Methods	B@1	B@2	B@3	B@4	M	R	C	S_m
VLAD-RNN [8]	63.1	51.9	46.1	42.1	29.7	58.8	200.7	82.8
VLAD-LSTM [8]	70.2	60.9	55.0	50.3	34.6	65.2	231.3	95.4
Soft Attention [8]	74.5	65.5	58.6	52.5	38.9	72.4	261.2	106.3
Hard Attention [8]	81.6	73.1	67.0	61.8	42.6	77.0	299.5	120.2
FC-ATT+LSTM [9]	81.4	75.0	68.5	63.5	41.7	75.0	299.6	120.0
SM-ATT+LSTM [9]	81.5	75.8	69.4	64.6	42.4	76.3	318.6	125.5
Sound-A-A [48]	74.8	68.4	63.1	59.0	36.2	65.8	272.8	108.5
TCE-Loss Network [34]	82.1	76.2	71.4	67.0	47.8	75.7	285.5	119.0
Word-sentence framework [1]	79.3	72.4	66.7	62.0	44.0	71.3	278.7	114.0
Recurrent-ATT [15]	85.2	79.3	74.3	69.8	45.7	80.7	338.9	133.8
GWFGA+LSGA [29]	83.2	76.6	71.0	66.0	44.4	78.5	332.7	130.4
SVM-D BOW [49]	76.4	66.6	58.7	52.0	36.5	68.8	271.4	107.2
SVM-D CONC [49]	76.5	69.5	64.2	59.4	37.0	68.8	292.3	114.4
Structured Attention [30]	85.4	80.4	75.7	71.5	46.3	81.4	334.9	133.5
MLCA-Net [5]	82.6	77.0	71.7	66.8	43.5	77.2	324.0	127.9
JTTS [17]	87.0	82.2	77.9	73.8	49.1	83.6	371.0	144.4
HCNet [3]	88.3	83.4	78.9	74.5	48.7	83.69	351.8	139.7
MVP [2]	89.1	84.2	79.9	75.9	49.0	84.4	379.1	147.1
Ours	**89.9**	**86.0**	**82.5**	**78.9**	**49.9**	**86.4**	**383.3**	**149.6**

TABLE IX
EXPERIMENTAL RESULTS ON RSICD DATASET

Methods	B@1	B@2	B@3	B@4	M	R	C	S_m
VLAD-RNN [8]	49.4	30.9	22.1	16.8	20.0	42.4	103.9	45.8
VLAD-LSTM [8]	50.0	32.0	23.2	17.8	20.5	43.3	118.0	49.9
Soft Attention [8]	67.5	53.1	43.3	36.2	32.6	61.1	196.4	81.6
Hard Attention [8]	66.7	51.8	41.6	34.1	32.0	60.8	179.3	76.6
FC-ATT+LSTM [9]	74.6	62.5	53.4	45.7	34.0	63.3	236.6	94.9
SM-ATT+LSTM [9]	75.7	63.4	53.9	46.1	35.1	64.6	235.6	95.4
Sound-A-A [48]	62.0	48.2	39.0	32.0	27.3	51.4	163.9	68.7
TCE-Loss Network [34]	76.1	63.6	54.7	47.9	34.3	66.9	246.7	99.0
Word-sentence framework [1]	72.4	58.6	49.3	42.5	32.0	62.6	206.3	85.9
Recurrent-ATT [15]	77.3	66.5	57.8	50.6	36.3	66.9	275.5	107.3
GWFGA+LSGA [29]	67.8	56.0	47.8	41.7	32.9	59.3	260.1	98.5
SVM-D BOW [49]	61.1	42.8	31.5	24.1	23.0	45.9	68.3	40.3
SVM-D CONC [49]	60.0	43.5	33.6	26.9	23.0	45.6	68.5	41.0
MLAT [50]	66.9	51.1	41.1	34.2	27.3	50.6	94.3	51.6
Structured Attention [30]	70.2	56.1	46.5	39.3	32.9	57.1	170.3	74.9
MLCA-Net [5]	75.7	63.4	53.9	46.1	35.1	64.6	235.6	95.4
JTTS [17]	78.9	68.0	58.9	51.4	37.7	68.2	279.6	109.2
MVP [2]	79.8	69.6	61.0	53.7	37.3	69.2	292.5	113.2
Ours	**82.9**	**72.8**	**64.2**	**57.0**	**40.8**	**72.1**	**312.5**	**120.6**

the CLIP encoder might lead to over-fitting on our specific task or data. In our final method, we freeze all parameters in pretrained CLIP.

E. Quantitative Results

To evaluate the effectiveness of our proposed method, we compare our method with existing several state-of-art methods, including VLAD-RNN [8], VLAD-LSTM [8], Soft Attention [8], Hard Attention [8], FC-ATT + LSTM [9], SM-ATT + LSTM [9], Sound-A-A [48], TCE-Loss Network [34], TCE-Loss Network [34], Word-sentence framework [1], Recurrent-ATT [15], GWFGA + LSGA [29], SVM-D BOW [49], SVM-D CONC [49], Structured Attention [30], MLCA-Net [5], JTTS [17], HCNet [3], MLAT [50], and MVP [2], on three challenging datasets. The experimental results are summarized as follows.

Table VIII shows the experimental results on one of the earliest RSIC datasets, UCM-Captions [7]. Although existing methods have achieved outstanding performance on it, our proposed method still gets the best scores on all metrics. Even if compared with the latest caption method [2], [3], [17], the mean language score S_m arrives 149.6%, which achieves a huge improvement.

Table IX reports the experimental results on the most popular RSIC dataset, RSICD [8]. Our method achieves the best performance for all metrics. Especially for CIDEr and mean language score S_m, it achieves a pretty significant improvement than all other methods. Compared with MLCA-Net [5], JTTS [17], and MVP [2], the improvements on RSICD are more obvious than the improvements on UCM-Captions. The main reason might be that RSICD contains more scene classes and more complex captions.

For one of the latest and most challenging NWPU-Captions datasets [5], Table X shows the outstanding performance of our method. Similar to RSICD, NWPU-Captions contains more scene classes and more complex captions, and our method also achieves good performance on it. The CIDEr and S_m achieve 214.1% and 104.1%, respectively. The outstanding performance on NWPU-Captions also proves the ability of our method for more challenging datasets.

We analyze the above experimental results. Specifically, MLAT [50] designs the multilayer aggregated transformer structure to efficiently utilize the features from different transformer encoding layers, which achieves high-quality feature representation. MLCA-Net [5] focuses on contextual information and multilevel visual features from the perspective of spatial and channel attention. JTTS [17] achieves the joint training for caption generation and scene classification, and implicitly adds the class information in the decoding stage. They implicitly use fine-grained visual features and scene class information to generate captions, and do not build explicit cross-modal prior information to assist in the cross-modal transformation, which will limit the important influence on caption generation. MVP [2] focuses on the cross-modal alignment ability of CLIP and uses it to help caption generation. It even uses multiple pretrained feature extractors to improve the visual semantic representation. Since it does not think about how to utilize the alignment ability and robust visual representation for cross-modal transformation, the capabilities of CLIP are not fully utilized.

Different from these methods, our method builds explicit scene prompts and entity prompts as prior text information based on the predicted results, which can provide more effective knowledge for caption generation. And, we achieve cross-modal semantic interaction between text prompts and fine-grained visual features to build a common semantic space, which can be regarded as an intermediate cross-modal semantic connection for decoding. Finally, the specific-designed attention fusion module considers the influence of dual prompts, visual features, and generated words, and dynamically fuses them together to generate a caption. Thus, our method can achieve a more significant improvement on more challenging datasets with complex annotations.

In all, for three RSIC datasets, we achieve better performance than existing methods on almost all metrics. Especially, it can be seen that our method shows more significant performance on the latest and more challenging RSICD and

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TABLE X

EXPERIMENTAL RESULTS ON NWPU-CAPTIONS DATASET. (* REPRESENTS THE REPRODUCED EXPERIMENTAL RESULTS BY US)

Methods	B@1	B@2	B@3	B@4	M	R	C	S_m
Soft Attention [8]	73.1	60.9	52.5	46.2	33.9	59.9	113.6	63.4
Hard Attention [8]	73.3	61.0	52.7	46.4	34.0	60.0	110.3	62.7
FC-ATT+LSTM [9]	73.6	61.5	53.2	46.9	33.8	60.0	123.1	66.0
SM-ATT+LSTM [9]	73.9	61.7	53.2	46.8	33.0	59.3	123.6	65.7
MLCA-Net [5]	74.5	62.4	54.1	47.8	33.7	60.1	126.4	67.0
MLAT* [50]	88.9	80.6	73.7	68.0	44.3	77.4	199.2	97.2
HCNet [3]	89.5	82.5	76.6	71.7	**47.4**	80.5	209.3	102.2
Ours	**90.9**	**84.2**	**78.6**	**73.8**	47.2	**81.4**	**214.1**	**104.1**

TABLE XI

EXPERIMENTAL RESULTS ON VRSBENCH DATASET

Method	B@1	B@2	B@3	B@4	M	R	C	S_m
Multi-modal Large Language Models (MLLMs)								
GeoChat w/o ft [51]	13.9	6.6	3.0	1.4	7.8	13.2	0.4	5.7
GPT-4V [52]	37.2	22.5	13.7	8.6	20.9	30.1	19.1	18.9
MiniGPT-v2 [53]	36.8	22.4	13.9	8.7	17.1	30.8	21.4	19.5
LLaVA-v1.5 [54]	**48.1**	31.5	21.2	**14.7**	**21.9**	36.9	33.9	26.9
GeoChat [51]	46.7	30.2	20.1	13.8	21.1	35.2	28.2	24.6
Other Caption Method								
MLAT [50]	34.7	17.8	9.2	4.9	12.7	26.3	3.1	11.8
Ours	47.7	**31.6**	**21.4**	**14.7**	21.0	**37.8**	**34.8**	**27.1**

NWPU-Captions datasets, which fully illustrates that our proposed method can better analyze multiple classes of RSI scenes and generate captions with higher quality and more details. Table XI reports the experimental results on the VRSBench [40], which contains more detailed and complex caption annotations. Due to the absence of scene class annotations in the VRSBench dataset, we have to remove the modules in our method that are related to scene prediction and scene prompt. Consequently, our method is reduced from utilizing both scene and entity prompts to relying solely on the entity prompt. In Table X, GeoChat w/o ft [51] means the results without fine-tuning, and the suboptimal results indicate that it struggles to adapt directly to the RSIC task without fine-tuning. It can be seen that our method remains competitive on almost all metrics when compared to the MLLMs, such as MiniGPT [53], GPT-4V [52], and GeoChat [51]. Even compared to more robust models like LLaVA-v1.5, our method still achieves the best performance in the metrics of BLEU@4, ROUGE_L, and CIDEr. And the mean language score S_m of our method also surpasses all comparative methods, demonstrating consistent superiority across multiple evaluation metrics.

The results show the potential benefits of our method in handling the detailed RSIC. By only building on entity prompt and cross-modal interaction, our method also achieves high-quality results even in the absence of scene class annotations. It is widely recognized that fine-tuning MLLMs is often a challenging and resource-intensive process due to the substantial computational requirements and costs. In contrast, our method has some competitive performance not only in terms of the experimental results but also independent of MLLMs. This makes our method practical for applications with limited computing resources.

TABLE XII

SOME EXPERIMENTAL RESULTS OF METHODS WITH MLLMS ON NWPU-CAPTIONS AND RSICD DATASET; ⋆ MEANS THE ZERO-SHOT CAPTION GENERATION SETTING, AND ∘ MEANS THE SUPERVISED CAPTION GENERATION SETTING

Dataset	Model	B@1	B@2	B@3	B@4	M	R	C	S_m
NWPU-Captions	GeoChat⋆ [51]	32.5	13.5	5.0	1.8	11.7	25.7	13.5	13.2
	MiniGPT-v2⋆ [53]	38.3	17.7	7.7	3.0	10.4	24.4	18.6	14.1
	LLaVA-v1.5⋆ [54]	45.4	23.3	11.7	5.7	14.1	28.3	28.5	19.2
	RS-CapRet∘ [55]	87.1	78.6	71.3	65.0	43.9	77.5	191.9	94.6
	Ours⋆	**49.4**	**27.4**	**15.0**	**7.9**	**16.3**	**31.5**	**46.6**	**25.6**
	Ours∘	**90.9**	**84.2**	**78.6**	**73.8**	**47.2**	**81.4**	**214.1**	**104.1**
RSICD	GeoChat⋆ [51]	22.9	9.1	3.3	1.3	10.9	21.0	11.5	11.2
	MiniGPT-v2⋆ [53]	32.5	15.9	7.6	3.2	10.8	22.7	23.3	15.0
	LLaVA-v1.5⋆ [54]	34.0	17.0	8.9	4.3	14.4	25.4	31.2	18.0
	RS-CapRet∘ [55]	74.1	62.2	52.9	45.5	37.6	64.9	260.5	102.1
	Ours⋆	**38.5**	**21.1**	**11.6**	**6.1**	**16.4**	**27.5**	**49.2**	**24.8**
	Ours∘	**82.9**	**72.8**	**64.2**	**57.0**	**40.8**	**72.1**	**312.5**	**120.6**

In addition, we also make comparison experiments with MLLMs like GeoChat [51], MiniGPT-v2 [53], and LLaVA-v1.5 [54] in the zero-shot caption generation setting based on the prompt "Please describe the image in one sentence," which is shown in Table XII. Since our method contains modules with random initialization, our method cannot directly achieve zero-shot description generation without any fine-tuning. Thus, we directly merge LLaVA-v1.5 into our method as a decoder, and sent our core idea of dual prompts into it to achieve zero-shot caption generation. Specifically, we construct dual prompts "The image about $\{\mathcal{C}\}$ might contains $\{\nu_1\}$, $\{\nu_2\}$,.... Please describe the image in one sentence." based on the CLIP model and add it into LLaVA-v1.5, which can provide additional semantic context as a cross-modal bridge that helps the model better understand the RSIs. It can be seen that while these MLLMs have shown progress in zero-shot caption generation, there is still a significant gap compared to supervised methods. When we modified our method into a zero-shot setting, compared with LLaVA-v1.5, the performance on two typical datasets achieves an obvious improvement. Additionally, we also show the performance of RS-CapRet [55], a supervised method based on CLIP + LLMs, to further illustrate the advantages of our method under supervised conditions.

F. Qualitative Results

To demonstrate the effectiveness of our method more intuitively, we show some successful qualitative results of different scenes for RSIC in Fig. 4. In addition, we also show the proposed two kinds of scene and entity prompts, which can better show the effective function of prompt prior information. "GT" means one of the five caption GT annotations in datasets, "SceP" means the predicted scene class prompt (blue color), and "EntP" means the explicit entity concept prompt (green color). "Ours" means the generated caption of our final method.

As shown in Fig. 4, we use red color to highlight some accurate and appropriate words, which indicates that the output of our method can accurately describe key information in RSIs. Even for difficult images containing multiple objects or having dim scenes, our method still performs well. For example, in Fig. 4(a), the terminal can be successfully described. In

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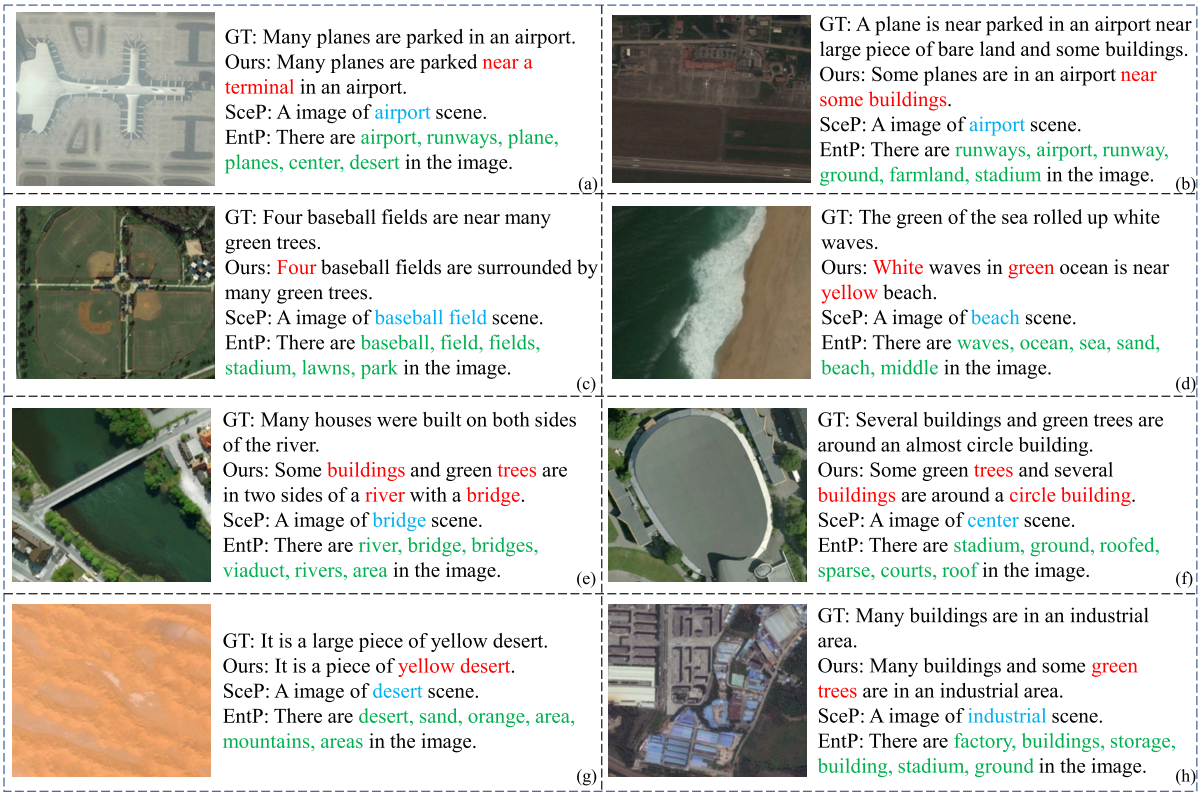


Fig. 4. Visualization of RSIC with different scenes. "GT" means the ground-truth caption, "SceP" means the predicted scene class prompt (blue color), and "EntP" means the explicit entity concept prompt (green color).

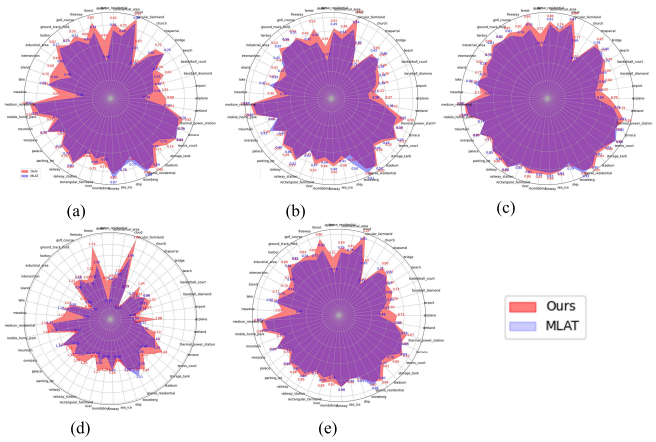


Fig. 5. Radar maps of different metrics with multiple scene classes of our method and MLAT method on NWPU-Captions. (a) BLEU@4. (b) METEOR. (c) ROUGE$_L$. (d) CIDEr. (e) S_m.

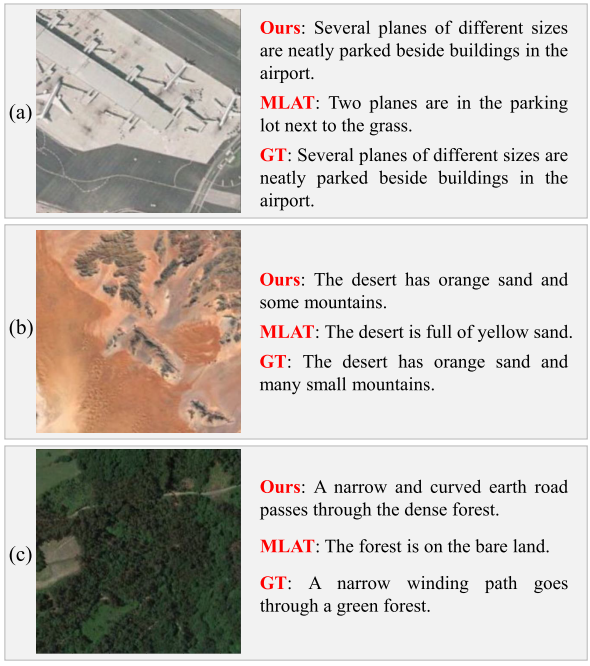


Fig. 6. Visualization of our method and MLAT method with different scenes in NWPU-Captions. "GT" means the ground-truth caption.

Fig. 4(b), our method can recognize the buildings in dim scenes. In Fig. 4(c), (d), and (g), the details such as number and color can all be accurately described. The images of Fig. 4(e) and (f) contain multiple objects with different scales, and the image in Fig. 4(h) contains hard objects such as the tree in the corner, which are all successfully described. In addition, it can also be seen that the scene classes of all images are accurately predicted, and most entity concepts match the image content, which can provide more prior information to help generate detailed and accurate descriptions. The above visualization results more intuitively demonstrate that most of the core words are contained in the prompts and the generated captions are better, which also shows the effectiveness of our proposed method.

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In addition, we also summarize the metrics for our method and MLAT [50] across all scene classes of the NWPU-Captions dataset to provide a more comprehensive analysis of the performance gains. Specifically, we use the radar map to show the performance gains in all scene classes. As shown in Fig. 5, our method achieves a substantial improvement in almost all scene classes. Especially in the classes of "airplane," "desert," and "forest," the CIDEr score of our method is 54.4%, 53.3%, and 43.6% higher than that of the MLAT method, respectively.

We further show more qualitative comparisons with the MLAT method [50] and our proposed method, as shown in Fig. 6. As shown in Fig. 6, the generated captions of MLAT are almost correspond to the input images. Compared with it, our method has more advantages on details and key elements. In Fig. 6(a), our method accurately captures the details of "different sizes" and "neatly parked beside buildings," which closely aligns with GT. In contrast, MLAT misses the variety in plane sizes and the proximity to buildings. In Fig. 6(c), our method describes "a narrow and curved Earth road passes through the dense forest," which is also more aligned with "a narrow winding path goes through a green forest" in GT. These results also highlight the strengths of our method in generating more detailed and accurate captions, especially in capturing key elements of the scenes.

V. Conclusion

In this article, we propose a novel dual-prompt-aware cross-modal semantic interaction and fusion network to achieve RSIC, which aims to utilize the ability of cross-modal alignment in VLM to help accurate caption generation. It builds dual scene and entity prompts to achieve cross-modal interaction and fusion as a cross-modal semantic connection, which can provide robust prior information for vision and text transformation. We propose an ECE to obtain various entity concepts for input images as the entity prompt. Next, we predict scene class and obtain fine-grained visual feature based on the MSP. Then, a PCI module is designed to build a cross-modal common semantic space as an intermediate connection to assist in caption generation. In the end, after achieving cross-modal semantic interaction, we design a prompt-aware attention fusion module for the transformer decoder to utilize cross-modal prompt feature to generate accurate captions. We conduct extensive experiments on the latest and challenging UCM-Captions, RSICD, and NWPU-Captions datasets, and our proposed method outperforms previous methods with a significant margin on almost all metrics.

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